

Digital Signal Processing

Revised Notes: Definitions and Proofs

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These notes cover the supplied lectures through inverse systems: signal representation, sampling and quantization, discrete-time systems, convolution, and correlation. Definitions, assumptions, proofs, and representative calculations are given together. Section 9 retains and develops the short difference-equation and FIR/IIR supplement in the attached compilation. The original lecture marks the midsemester cutoff immediately before FIR/IIR.

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Notation and conventions

$x_a(t), x[n]$	Continuous-time signal and discrete-time sequence; $n \in \mathbb{Z}$.
$\mathbb{C}, j$	Complex numbers and $j^2 = -1$; real sequences are a special case.
$x^*[n]$	Complex conjugate of $x[n]$.
T_s, f_s	Sampling interval and sampling rate, $f_s = 1/T_s$.
f, Ω, ω	Frequency in Hz, rad/s, and rad/sample: $\Omega = 2\pi f, \omega = \Omega T_s$.
$\delta[n], u[n]$	Discrete unit impulse and unit step, with $u[0] = 1$.
$S_r x[n] = x[n - r]$	Delay by r samples; $r < 0$ is an advance.
$T\{x\}[n]$	Output at index n of the system T driven by x .
$h[n], s[n]$	Impulse response $T\{\delta\}[n]$ and step response $T\{u\}[n]$.
$x * h, r_{xy}[\ell]$	Linear convolution and correlation with the convention in Section 7.

An underlined list entry denotes $n = 0$; unlisted entries of a finite sequence are zero. Infinite sums are used only where they converge; hypotheses allowing an interchange of sums or an infinite superposition are stated explicitly.

1 Signals and analog–digital conversion

1.1 What a signal represents

Definition 1.1 (Signal). A signal is a function $x : \mathcal{D} \rightarrow \mathcal{A}$ representing information about a physical or abstract quantity. The domain supplies the independent variable(s), and the codomain supplies the possible values. Examples are acoustic pressure versus time, image intensity versus two spatial coordinates, density versus depth, and temperature versus altitude.

A signal need not depend on time. A wave is a propagating disturbance that can carry a signal: mechanical waves require a material medium, while electromagnetic waves do not.

The independent variable and amplitude must be classified separately:

Domain	Continuous-valued amplitude	Discrete-valued amplitude
Continuous	Usual ideal analog waveform	Level-valued waveform, e.g. a switching signal
Discrete	Sampled but unquantized sequence	Digital sequence

Thus continuous-time does not mean that x_a is a continuous *function*, and sampling alone does not quantize amplitude. The term analog usually refers to continuous representation of a quantity. A digital representation uses a discrete alphabet, typically finite binary words.

Other classifications include one- versus multidimensional; real versus complex; natural versus synthetic; deterministic versus random; finite versus infinite duration; and periodic versus aperiodic. A deterministic signal is specified by a fixed rule; a random signal is modelled as a family of random variables indexed by time or another domain.

1.2 Why process signals digitally?

Processing extracts useful information, suppresses disturbances, or changes a representation. Examples include filtering, detection, estimation, modulation and demodulation, multiplexing and demultiplexing, compression, and control. In an ECG, for example, one may wish to preserve the waveform while suppressing electrical pickup and muscle-activity interference.

Advantages	Costs and limitations
Programmable, repeatable, adjustable processing; storage and easy replication	Sampling interfaces, conversion errors, latency, computation and power consumption
Low sensitivity to component tolerances; high precision with sufficient word length	Finite word length causes quantization, roundoff, overflow and finite dynamic range
Multirate processing, adaptive coefficients, and suitable filters with linear phase	Input bandwidth is limited by conversion speed and the sampling rate
Digital stages can be cascaded without analog loading between mathematical stages	Physical acquisition and reconstruction still require suitable analog circuitry

Time-division multiplexing interleaves different sources' samples into assigned time slots; a synchronized receiver separates them. DSP does not automatically remove all noise or make accuracy and dynamic range unlimited.

1.3 Sampling, holding, quantization and encoding

analog input $\rightarrow$ anti-alias filter $\rightarrow$ sample/hold $\rightarrow$ quantizer $\rightarrow$ encoder

Definition 1.2 (Uniform sampling). For $T_s > 0$, the samples of x_a are

$$x[n] = x_a(nT_s), \quad n \in \mathbb{Z}, \quad f_s = \frac{1}{T_s}. \quad (1)$$

An ideal sampler records instantaneous values. A sample-and-hold circuit holds a measured value approximately constant while conversion occurs. The held waveform is continuous-time, even though it is piecewise constant.

Definition 1.3 (Uniform mid-rise quantizer). Let the input range be $[-A, A]$, $A > 0$, and use B bits per sample. Set

$$L = 2^B, \quad \Delta = \frac{2A}{L}.$$

Divide the range into L bins of width Δ , and replace every sample in bin i by its midpoint

$$q_i = -A + (i + \tfrac{1}{2})\Delta, \quad i = 0, \dots, L-1.$$

Choose one fixed convention at bin boundaries and include A in the last bin. Encoding assigns a distinct B -bit word to each q_i .

Proposition 1.4 (Quantization error bound). If the input is within range, then $e_q = x_q - x$ satisfies

$$|e_q| \leq \frac{\Delta}{2}.$$

Proof. Every point of a bin is within half its width of the midpoint. This remains true at a boundary under either adjacent-bin choice. $\square$

For $B = 3$, the eight levels are $\{-7A/8, -5A/8, -3A/8, -A/8, A/8, 3A/8, 5A/8, 7A/8\}$, encoded from 000 to 111. If instead L equally spaced *levels include both* endpoints, their spacing is $2A/(L-1)$. These are different conventions. Saturation outside the specified range can violate the half-step error bound.

Proposition 1.5 (Idealized quantization-noise calculation). Under the approximation $e_q \sim \text{Unif}[-\Delta/2, \Delta/2]$,

$$\mathbb{E}e_q = 0, \quad \mathbb{E}e_q^2 = \frac{\Delta^2}{12}.$$

For a sinusoid with mean-square value $A^2/2$, this model gives

$$\text{SQNR}_{\text{dB}} = 10 \log_{10} \frac{A^2/2}{\Delta^2/12} = 10 \log_{10}(3/2) + 20B \log_{10} 2 \approx 1.76 + 6.02B.$$

Proof. Symmetry gives the mean, and $\Delta^{-1} \int_{-\Delta/2}^{\Delta/2} e^2 \, de = \Delta^2/12$. Substitute $\Delta = 2A/2^B$ into the ratio of signal power to noise power. $\square$

This is a high-resolution noise model, not a claim that every deterministic quantization error is uniform or independent of the signal. The sinusoid’s mean-square value is taken over its phase or a nondegenerate sampling pattern.

1.4 D/A conversion and reconstruction

digital words $\rightarrow$ decoded sample values $\rightarrow$ DAC/hold $\rightarrow$ reconstruction filter $\rightarrow$ analog output

A zero-order hold has the idealized output

$$x_{\text{hold}}(t) = \sum_{n \in \mathbb{Z}} x_q[n] \mathbf{1}_{[nT_s, (n+1)T_s)}(t).$$

A reconstruction low-pass filter suppresses the unwanted replicated spectral components of the sampled/hold signal. For a desired band $|f| \leq f_{\text{max}} < f_s/2$, an ideal cutoff may be chosen with $f_{\text{max}} < f_c < f_s - f_{\text{max}}$; choosing it between f_{max} and $f_s/2$ is common. Exact reconstruction also requires accounting for the hold’s amplitude/phase distortion; a hold followed by an arbitrary low-pass filter is not automatically the ideal sinc interpolator. Quantization error remains even with ideal subsequent reconstruction.

2 Discrete-time sequences and their algebra

2.1 Representation, support and complex values

A sequence is a function $x : \mathbb{Z} \rightarrow \mathbb{C}$. Formulae, tables, lists and stem plots are equivalent representations when the indexing is specified. For example,

$$x = \{1, -1, 2, 3\} \iff x[-2] = 1, x[-1] = -1, x[0] = 2, x[1] = 3,$$

with zero elsewhere. Its support is $\text{supp } x = \{n : x[n] \neq 0\}$. For a nonzero finite sequence whose first and last nonzero indices are a, b , its *span length* is $N = b - a + 1$. This includes any internal zero entries. A right-sided sequence has support bounded below; a left-sided sequence has support bounded above. A sequence extending without bound in both directions is two-sided. A finite sequence is both right- and left-sided.

Write $x[n] = x_R[n] + jx_I[n]$, with

$$|x[n]| = \sqrt{x_R[n]^2 + x_I[n]^2}, \quad \arg x[n] = \text{atan2}(x_I[n], x_R[n]) \quad (x[n] \neq 0).$$

Phase is undefined at zero. Real/imaginary parts or magnitude/phase may be plotted separately.

2.2 Pointwise operations and changes of index

Addition and multiplication are pointwise: $(x + y)[n] = x[n] + y[n]$ and $(xy)[n] = x[n]y[n]$. Pointwise multiplication is different from convolution.

Operation	Formula	Effect
Amplitude scaling	$ax[n]$	Multiply every value by a .
Delay by $r > 0$	$S_r x[n] = x[n - r]$	Move the sample at m to $m + r$.
Advance by $r > 0$	$S_{-r} x[n] = x[n + r]$	Move the sample at m to $m - r$.
Reversal	$Rx[n] = x[-n]$	Move the sample at m to $-m$.

Proposition 2.1 (Index mapping and order of operations). For $y[n] = x[g(n)]$, the input sample at m appears at the integer solutions of $g(n) = m$. In particular,

$$S_r S_s = S_{r+s}, \quad R^2 = I, \quad RS_r = S_{-r}R.$$

Proof. The value $x[m]$ is selected exactly when $g(n) = m$. Also, $S_r S_s x[n] = x[n - r - s]$, $R^2 x[n] = x[n]$, and $RS_r x[n] = x[-n - r] = S_{-r}Rx[n]$. $\square$

For $y[n] = x[1 - n]$, a sample at m moves to $1 - m$: reverse and then delay by one. For $x[2n - 1]$, only odd-indexed input samples occur, at $n = (m + 1)/2$. An expression $x[n/L]$ is undefined when $n/L \notin \mathbb{Z}$ until a rule, such as zero insertion, has been specified.

The unit delay is often drawn as z^{-1} ; here it denotes the operator $S_1 x[n] = x[n - 1]$. Cascading N such blocks gives $x[n - N]$.

2.3 Unit impulse, unit step and signal expansions

Definition 2.2. The discrete unit impulse and step are

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0, \end{cases} \quad u[n] = \begin{cases} 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

The discrete impulse is a sequence, not the continuous-time Dirac distribution.

Proposition 2.3 (Elementary identities).

$$\begin{aligned} u[n] &= \sum_{k=0}^{\infty} \delta[n-k], & \delta[n] &= u[n] - u[n-1] = u[n]u[-n], \\ u[n] + u[-n] &= 1 + \delta[n], & u[n] + u[-n-1] &= 1, \\ u[n] - u[-n] &= \text{sgn}(n), & u[n] - u[-n-1] &= 2u[n] - 1. \end{aligned}$$

Here $\text{sgn}(0) = 0$. For integers $a \leq b$,

$$\mathbf{1}_{\{a \leq n \leq b\}} = u[n-a] - u[n-b-1].$$

Proof. In the first sum, the sole potentially nonzero term has $k = n$, and it is present exactly when $n \geq 0$. The other identities follow by checking $n < 0, n = 0, n > 0$, or, for the window, $n < a, a \leq n \leq b$, and $n > b$. $\square$

Theorem 2.4 (Impulse and step expansions). Every sequence has the pointwise identity

$$x[n] = \sum_{k \in \mathbb{Z}} x[k] \delta[n-k]. \quad (2)$$

If $\lim_{k \rightarrow -\infty} x[k] = c$ exists, then

$$x[n] = c + \sum_{k=-\infty}^n (x[k] - x[k-1]) = c + \sum_{k \in \mathbb{Z}} (x[k] - x[k-1]) u[n-k]. \quad (3)$$

The sums in (3) are interpreted as limits of lower-truncated sums.

Proof. Only $k = n$ contributes in (2). For $a \leq n$, telescoping gives $\sum_{k=a}^n (x[k] - x[k-1]) = x[n] - x[a-1]$. Let $a \rightarrow -\infty$. The step factor simply enforces $k \leq n$. $\square$

For $x = \{2, 4, 1\}$,

$$\begin{aligned} x[n] &= 2\delta[n] + 4\delta[n-1] + \delta[n-2] \\ &= 2u[n] + 2u[n-1] - 3u[n-2] - u[n-3]. \end{aligned}$$

The final $-u[n-3]$ returns the signal to zero. The impulse expansion is valid pointwise for every sequence; moving its infinite sum through a system requires an additional justification (Section 6).

2.4 Even and odd decomposition

Definition 2.5. A sequence is even if $x[-n] = x[n]$, and odd if $x[-n] = -x[n]$. These definitions also apply to complex sequences and do not involve conjugation.

Theorem 2.6 (Unique parity decomposition). Every sequence decomposes uniquely as $x = x_e + x_o$, where

$$x_e[n] = \frac{x[n] + x[-n]}{2}, \quad x_o[n] = \frac{x[n] - x[-n]}{2}.$$

Proof. Replacing n by $-n$ proves the asserted parities, and adding the expressions recovers $x[n]$. If also $x = a + b$ with a even and b odd, then $x[-n] = a[n] - b[n]$. Adding and subtracting these two equations forces the displayed formulae. In particular an odd sequence has $x_o[0] = 0$. $\square$

2.5 Exponential signals and periodicity

A real exponential has the form Aa^n wherever defined; a complex exponential can be written $Ar^n e^{j\omega n}$, $r > 0$. For a right-sided exponential, $r < 1$ gives decay and $r > 1$ growth. A sinusoid is $A \cos(\omega n + \phi)$, and $e^{j\theta} = \cos \theta + j \sin \theta$.

Definition 2.7. A positive integer N is a period of x if $x[n + N] = x[n]$ for every n . A periodic sequence has a least positive integer period N_0 , its fundamental period. Every constant sequence, including the zero sequence, has $N_0 = 1$.

Theorem 2.8 (Discrete sinusoidal periodicity). For $A \neq 0$, the sequence $Ae^{j(\omega_0 n + \phi)}$ is periodic exactly when $\omega_0/(2\pi) \in \mathbb{Q}$. If $\omega_0/(2\pi) = p/q$ in lowest terms, $q > 0$, its fundamental period is q .

Proof. The ratio of the samples at $n + N$ and n is $e^{j\omega_0 N}$. It equals one exactly when $\omega_0 N = 2\pi m$ for an integer m . For $\omega_0/(2\pi) = p/q$, this requires $q \mid N$, since p, q are coprime; the least choice is $N = q$. $\square$

For a nonzero cosine the same criterion and period hold, apart from an identically zero sampled sinusoid. To see necessity when $\omega_0 \notin \pi\mathbb{Z}$, expand $\cos(\omega_0(n + N) + \phi) - \cos(\omega_0 n + \phi)$ into the two independent sequences $e^{j\omega_0 n}$ and $e^{-j\omega_0 n}$. Their coefficients must vanish, giving $e^{j\omega_0 N} = 1$. Independence follows by evaluating a linear combination at $n = 0, 1$, whose determinant is nonzero when $\sin \omega_0 \neq 0$. If $\omega_0 \in \pi\mathbb{Z}$, direct substitution gives a constant, an alternating sequence, or the zero sequence. For example, $\sin(\pi n) \equiv 0$.

Thus $\cos(3n)$ is aperiodic, while $\cos(3\pi n/7)$ has period 14. Unlike continuous time, a discrete period must be an integer number of samples.

Proposition 2.9 (Combining periodic sequences). If $x_1, \dots, x_r$ have periods $N_1, \dots, N_r$, then $N = \text{lcm}(N_1, \dots, N_r)$ is a period of their sum and product. It need not be the fundamental period of either result.

Proof. Each N/N_i is an integer, so $x_i[n + N] = x_i[n]$ for every i . Adding or multiplying preserves the equality. Cancellation can shorten the period: $(-1)^n + [-(-1)^n] = 0$ has fundamental period one although both components have fundamental period two. $\square$

Precision check. When applying the rational-frequency criterion, reduce the fraction fully. The continuous-time formula $2\pi/\omega_0$ is not automatically an integer or a valid discrete fundamental period.

3 Boundedness, summability, energy and power

3.1 Definitions and relations between signal classes

Definition 3.1. For a sequence x , define

$$\begin{aligned} \|x\|_\infty &= \sup_{n \in \mathbb{Z}} |x[n]|, & \ell^\infty &= \{x : \|x\|_\infty < \infty\}, \\ \|x\|_1 &= \sum_{n \in \mathbb{Z}} |x[n]|, & \ell^1 &= \{x : \|x\|_1 < \infty\}, \\ E_x &= \|x\|_2^2 = \sum_{n \in \mathbb{Z}} |x[n]|^2, & \ell^2 &= \{x : E_x < \infty\}. \end{aligned}$$

These are the bounded, absolutely summable, and square-summable sequences. When the limit exists, the symmetric time-average power is

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2. \quad (4)$$

An energy signal has $0 < E_x < \infty$; a power signal has $0 < P_x < \infty$. The zero sequence is excluded from these two labels by convention.

For continuous time the analogous quantities replace sums by integrals: $E_a = \int_{\mathbb{R}} |x_a(t)|^2 dt$ and $P_a = \lim_{T \rightarrow \infty} (2T)^{-1} \int_{-T}^T |x_a(t)|^2 dt$, if defined.

Theorem 3.2 (Strict inclusions and energy–power relation).

$$\ell^1 \subsetneq \ell^2 \subsetneq \ell^\infty.$$

Every finite-energy sequence has zero average power. Every power signal has infinite energy. Consequently, a nonzero signal cannot be both an energy signal and a power signal; some signals are neither.

Proof. If $x \in \ell^1$, then $\|x\|_\infty \leq \|x\|_1$, so

$$\sum_n |x[n]|^2 \leq \|x\|_\infty \sum_n |x[n]| < \infty.$$

If $x \in \ell^2$, each $|x[n]|^2 \leq E_x$, proving boundedness. The sequence $x[n] = 1/n$ for $n \geq 1$ and zero otherwise belongs to ℓ^2 but not ℓ^1 ; $x[n] = 1$ is bounded but not square summable. Finally,

$$0 \leq \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2 \leq \frac{E_x}{2K+1} \rightarrow 0$$

when $E_x < \infty$. A positive limiting power therefore precludes finite energy. $\square$

Neither the existence nor finiteness of average power follows merely from a signal being defined. A bounded sequence need not have a convergent power average.

3.2 Power of periodic sequences

Proposition 3.3. If x is N -periodic, then for every integer a ,

$$P_x = \frac{1}{N} \sum_{n=a}^{a+N-1} |x[n]|^2. \quad (5)$$

A nonzero periodic sequence has positive finite power and infinite energy.

Proof. Let $S = \sum_{n=a}^{a+N-1} |x[n]|^2$; periodicity makes this independent of a . Partition a window of $2K+1 = qN + r$ consecutive samples into q complete blocks and a remainder of $0 \leq r < N$ samples. Its energy is $qS + R_K$, where R_K is bounded independently of K . Dividing by $2K+1$ gives S/N in the limit. If $x \neq 0$, then $S > 0$, and infinitely many complete blocks make the total energy infinite. $\square$

For $x[n] = Ae^{j\omega n}$, $P_x = |A|^2$, whether or not the sequence is periodic. For $x[n] = A \cos(\omega n + \phi)$, with real A ,

$$P_x = \begin{cases} A^2/2, & \omega \notin \pi\mathbb{Z}, \\ A^2 \cos^2 \phi, & \omega \in \pi\mathbb{Z}. \end{cases}$$

Indeed, $\cos^2 \theta = (1 + \cos 2\theta)/2$, and for $e^{j2\omega} \neq 1$ the finite geometric sum of $e^{j2\omega n}$ is bounded independently of its length, so its average tends to zero.

3.3 Representative calculations

Sequence	Bounded	ℓ^1	Energy	Power
$\delta[n]$	yes	yes	1	0
a^n for $n \geq 0$, zero otherwise; $ a < 1$	yes	yes	$1/(1 - a ^2)$	0
$1/n$ for $n \geq 1$, zero otherwise	yes	no	finite	0
$u[n]$	yes	no	∞	$1/2$
1 for every n	yes	no	∞	1
$nu[n]$	no	no	∞	∞

The exponential energy is the geometric sum $\sum_{n=0}^{\infty} |a|^{2n}$. For the step, the numerator in (4) is $K + 1$, giving $P_u = 1/2$. For the ramp it is $K(K + 1)(2K + 1)/6$, giving average $K(K + 1)/6 \rightarrow \infty$. At $a = 0$, define the exponential sequence as $\delta[n]$.

Precision check. With the one-sided average $(K + 1)^{-1} \sum_{n=0}^K |x[n]|^2$, the step has power one. Both answers reflect their stated conventions. Signal boundedness and a system's BIBO stability are different properties.

4 Sampling, aliasing and rate conversion

4.1 Analog and digital frequency

Sampling $x_a(t) = A \cos(2\pi f t + \phi)$ at $t = nT_s$ gives

$$x[n] = A \cos(\omega n + \phi), \quad \boxed{\omega = 2\pi f T_s = \frac{2\pi f}{f_s} = \Omega T_s.}$$

Analog frequencies f and Ω have units Hz and rad/s; ω has units rad/sample. The normalized frequency in cycles/sample is f/f_s .

Proposition 4.1 (Frequency equivalence). Two unit complex exponentials $e^{j\omega_1 n}$, $e^{j\omega_2 n}$ give the same sequence exactly when $\omega_2 - \omega_1 \in 2\pi\mathbb{Z}$. Two zero-phase unit cosines give the same sequence exactly when $\omega_2 = \pm\omega_1 + 2\pi k$ for an integer k .

Proof. For exponentials, equality at $n = 1$ requires $e^{j(\omega_2 - \omega_1)} = 1$, and this condition gives equality at every integer n . For cosines, equality at $n = 1$ implies $\cos \omega_2 = \cos \omega_1$, whose solutions are $\omega_2 = \pm\omega_1 + 2\pi k$. Evenness and 2π -periodicity of cosine give sufficiency for all n . $\square$

One half-open interval of length 2π , such as $(-\pi, \pi]$, represents all distinct complex-exponential frequencies. For zero-phase cosines one can further fold to $[0, \pi]$. With phase, reversing the frequency reverses the phase:

$$\cos(-\omega n + \phi) = \cos(\omega n - \phi).$$

Digital frequency is periodic as a *parameter*; this does not mean that every sinusoidal sequence is periodic in n .

4.2 Aliasing and the sampling-rate condition

Definition 4.2 (Aliasing). Aliasing is the failure of distinct admissible continuous-time signals to remain distinguishable after sampling. In particular, sampled complex exponentials alias when $f_2 = f_1 + kf_s$, and zero-phase real cosines alias when

$$\boxed{f_2 = \pm f_1 + kf_s, \quad k \in \mathbb{Z}.} \quad (6)$$

This is the preceding proposition after substituting $\omega_i = 2\pi f_i/f_s$. To find the observed frequency of a real zero-phase tone, subtract an integer multiple of f_s to put it in $[-f_s/2, f_s/2]$, then take the absolute value.

Example 4.3 (Three analog tones with identical samples). The signals $\cos(6\pi t)$, $\cos(14\pi t)$, $\cos(26\pi t)$ have frequencies 3, 7, 13 Hz. At $f_s = 10$ Hz their digital frequencies are 0.6π , 1.4π , 2.6π , and

$$\cos(1.4\pi n) = \cos((2\pi - 0.6\pi)n) = \cos(0.6\pi n), \quad \cos(2.6\pi n) = \cos(0.6\pi n).$$

At $f_s = 50$ Hz, the digital frequencies 0.12π , 0.28π , 0.52π are distinct, and $50 > 2 \cdot 13$.

Definition 4.4 (Band limitation and Nyquist terminology). A baseband signal is band-limited to $f_{\max}$ if its spectral content vanishes outside $[-f_{\max}, f_{\max}]$. The *Nyquist rate* is $2f_{\max}$, determined by the signal band; the *Nyquist frequency* is $f_s/2$, determined by the chosen sampling rate. A robust sufficient condition for ideal recovery is

$$f_s > 2f_{\max}.$$

An analog anti-aliasing filter must suppress unwanted frequencies *before* sampling. Once two signals have exactly the same samples, no operation on those samples alone can identify which signal was present. Indeed, any deterministic reconstruction rule receives the same input and must give the same output for both. Extra prior information can restrict the admissible signal class.

4.3 Ideal interpolation: a precise theorem and proof

The next proof uses only a local frequency representation and the standard completeness of Fourier exponentials on a finite interval. No DFT or Z-transform machinery is needed.

For a precise finite-energy version of band limitation, suppose

$$x_a(t) = \int_{-f_{\max}}^{f_{\max}} F(f) e^{j2\pi f t} df, \quad F \in L^2([-f_{\max}, f_{\max}]). \quad (7)$$

Here F is the spectral amplitude, extended by zero outside the band. Since the interval is finite, Cauchy–Schwarz implies $F \in L^1$, so this formula defines a continuous signal and its point samples unambiguously.

Theorem 4.5 (Sampling and sinc reconstruction). Under (7), if $f_s = 1/T_s > 2f_{\max}$, then

$$x_a(t) = \sum_{n \in \mathbb{Z}} x_a(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right), \quad (8)$$

where $\operatorname{sinc}(v) = \sin(\pi v)/(\pi v)$ for $v \neq 0$, and $\operatorname{sinc}(0) = 1$. The symmetric partial sums converge uniformly in t . The samples determine x_a uniquely within this band-limited class.

Proof. Extend F by zero to $I = [-f_s/2, f_s/2]$. Expand it in the complete orthogonal basis $\{e^{-j2\pi f n T_s} : n \in \mathbb{Z}\}$ of $L^2(I)$. Its coefficients are

$$c_n = \frac{1}{f_s} \int_I F(f) e^{j2\pi f n T_s} df = T_s x_a(nT_s).$$

Thus $F_N(f) = T_s \sum_{n=-N}^N x_a(nT_s) e^{-j2\pi f n T_s}$ tends to F in $L^2(I)$. Inverting each finite sum gives

$$\begin{aligned} \int_I F_N(f) e^{j2\pi f t} df &= T_s \sum_{n=-N}^N x_a(nT_s) \int_{-f_s/2}^{f_s/2} e^{j2\pi f(t - nT_s)} df \\ &= \sum_{n=-N}^N x_a(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right). \end{aligned}$$

The inverse integral of the error is bounded, independently of t , by

$$\left| \int_I (F - F_N)(f) e^{j2\pi ft} df \right| \leq \sqrt{f_s} \|F - F_N\|_{L^2(I)} \longrightarrow 0.$$

This proves (8) and uniform convergence. If all samples vanish, all c_n vanish; completeness gives $F = 0$ almost everywhere and $x_a = 0$. Applying this to the difference of two candidate signals proves uniqueness. $\square$

At $t = mT_s$, the sinc factors are $\delta[m - n]$, so reconstruction agrees with each sample. Between samples they specify the unique band-limited interpolant. Ideal sinc interpolation is nonlocal and generally noncausal; practical systems use finite approximations and accept delay and approximation error.

Precision check. For this finite-energy theorem, equality $f_s = 2f_{\max}$ also works: individual spectral endpoints have measure zero. For a class that includes pure tones with arbitrary phase at the band edge, equality is unsafe: $x_a(t) = \sin(\pi f_s t)$ is nonzero but $x_a(nT_s) = 0$ for every n . Such a pure tone is not in the finite-energy class (7). The strict inequality avoids this boundary ambiguity and allows a practical filter margin.

4.4 Upsampling and downsampling

Definition 4.6 (Integer rate changers). For integers $L, M \geq 1$, define

$$(U_L x)[n] = \begin{cases} x[n/L], & L \mid n, \\ 0, & L \nmid n, \end{cases} \quad (D_M x)[n] = x[Mn].$$

Upsampling inserts $L - 1$ zeros between samples; downsampling retains every M -th sample. The resulting physical sampling rates are Lf_s and f_s/M .

For $x = \{-1, -2, \underline{1}, 3, 4, 1, 5\}$, $D_2 x = \{-1, \underline{1}, 4, 5\}$. If $x = \{-1, \underline{1}, 2\}$, $U_3 x$ has values $-1, 1, 2$ at indices $-3, 0, 3$ and zero elsewhere.

Proposition 4.7 (Algebra and norms of rate changers). The operators U_L, D_M are linear, satisfy

$$\|U_L x\|_\infty = \|x\|_\infty, \quad \|D_M x\|_\infty \leq \|x\|_\infty,$$

and, for finite-energy sequences,

$$E_{U_L x} = E_x, \quad E_{D_M x} \leq E_x.$$

Their shift relations are

$$U_L S_r = S_{Lr} U_L, \quad D_M S_{Mr} = S_r D_M.$$

For $L, M > 1$, neither is time-invariant under the same-index shift test.

Proof. Selecting or inserting samples distributes over any linear combination. Upsampling preserves all input values and adds zeros; downsampling selects a subset. This proves the norm and energy statements. Direct substitution gives the shift identities. For non-invariance, both operators map δ to δ , but $U_L S_1 \delta = \delta[\cdot - L] \neq S_1 \delta$, and $D_M S_1 \delta = 0 \neq S_1 \delta$ when $M > 1$. $\square$

Filtering around rate changes. Zero insertion alone does not interpolate the missing amplitudes. An interpolation low-pass filter follows upsampling and removes the extra spectral images; its usual passband gain is L . Before downsampling, the retained band should be below the new Nyquist frequency $f_s/(2M)$. In input digital-frequency units this means $|\omega| < \pi/M$. Rational rate conversion by L/M uses upsampling, a low-pass filter, and then downsampling.

Example 4.8 (Linear interpolation by two). Set $v = U_2x$, then apply

$$w[n] = v[n] + \frac{1}{2}v[n-1] + \frac{1}{2}v[n+1].$$

At even indices, $w[2k] = x[k]$. At odd indices, $w[2k+1] = (x[k] + x[k+1])/2$, because the central zero is replaced by the average of its neighboring original samples. This explains the three-point operator in the lectures. It is linear interpolation, rather than exact sinc interpolation.

Index causality versus physical clocks. If input and output are treated as sequences on the same clock, D_M uses a future index at $n = 1$, and U_L uses a future index at $n = -L$: $(U_Lx)[-L] = x[-1]$. Thus neither is causal on all two-sided sequences under that shared-index definition when its factor exceeds one. In a physical multirate implementation the clocks differ: output index n occurs at nT_s/L for upsampling and nMT_s for downsampling. The copied input sample occurs at precisely that physical time; rate conversion itself does not require future physical data. A centered interpolation filter can introduce a separate look-ahead requirement.

5 Discrete-time systems: definitions and tests

5.1 Systems, admissible inputs and initial conditions

Definition 5.1 (System). A discrete-time system is a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ between specified classes of sequences. Its output is $y[n] = T\{x\}[n]$. Two systems are equal on $\mathcal{X}$ if their outputs agree for every $x \in \mathcal{X}$.

Admissible inputs and initial states are part of the model. Equality on one test input does not establish equality of two systems. A modelling error is $e[n] = y_{\text{physical}}[n] - y_{\text{model}}[n]$; a small error on one experiment alone gives no guarantee for other inputs.

Example 5.2 (Accumulator with zero prehistory). The accumulator is

$$(Ax)[n] = \sum_{k=-\infty}^n x[k]. \quad (9)$$

One convenient domain is the class $\mathcal{L}$ of sequences that vanish below some finite index (the index may depend on the sequence). This is a vector space closed under all integer shifts, and each sum in (9) is finite. The definition also extends to other inputs whose past sums converge. Subtracting adjacent sums gives

$$y[n] - y[n-1] = x[n].$$

For a causal input and $y[-1] = 0$, this becomes $y[n] = \sum_{k=0}^n x[k]$ for $n \geq 0$. Input $\{1, 2, 3, 4, 5\}$ gives output values 1, 3, 6, 10, 15, followed by 15 thereafter, rather than zero.

The recurrence by itself does not determine a unique output: any added constant solves the homogeneous equation. Fixing a nonzero input-independent initial value generally makes the input-output map affine rather than linear. A lower limit fixed at 0 for arbitrary two-sided inputs also fixes a time origin; it cannot silently replace the shift-invariant bilateral definition.

5.2 Linearity

Definition 5.3. On a vector space $\mathcal{X}$ over a specified field $\mathbb{K}$, the system T is linear if

$$T\{ax_1 + bx_2\} = aT\{x_1\} + bT\{x_2\} \quad \text{for all } x_1, x_2 \in \mathcal{X}, \ a, b \in \mathbb{K}. \quad (10)$$

This is equivalent to additivity and homogeneity. Here $\mathbb{K} = \mathbb{C}$ unless a real-input system is explicitly specified.

Proof of the equivalence. Set $a = b = 1$ for additivity, and $b = 0$ for homogeneity. Conversely, apply additivity to $ax_1 + bx_2$, then homogeneity to its two terms. $\square$

Linearity implies $T\{0\} = 0$, since homogeneity gives $T\{0x\} = 0T\{x\} = 0$. This is necessary but insufficient: $T\{x\}[n] = x[n]^2$ passes this check but $T\{2x\} = 4T\{x\}$, not $2T\{x\}$, for a nonzero input.

Proposition 5.4 (A useful class of linear operators). For fixed coefficients $c_i[n]$ and fixed integer-valued maps $g_i(n)$,

$$T\{x\}[n] = \sum_{i=1}^q c_i[n]x[g_i(n)]$$

is linear, irrespective of how nonlinear g_i looks as a function of n .

Proof. Replace x by $ax_1 + bx_2$, distribute the finite sum, and factor out a, b . The same proof applies to convergent sums when the corresponding operations are valid on the stated domain. $\square$

Thus $nx[n]$, $x[n^2]$, $x[-n]$, and the accumulator are linear. The systems $x[n]^2$, $x[n-1]x[n+1]$, $e^{x[n]}$, and $ax[n] + c$ with $c \neq 0$ are nonlinear. The first two fail homogeneity under amplitude scaling; the latter two fail $T\{0\} = 0$. Conjugation $x \mapsto x^*$ is real-linear but not complex-linear: $(jx)^* = -jx^*$.

5.3 Time invariance

Definition 5.5. On a domain closed under shifts, T is time-invariant if it commutes with every integer shift:

$$TS_r x = S_r T x \quad \text{for every } x \text{ and } r \in \mathbb{Z}. \quad (11)$$

Thus delaying the input by r delays the output by the same r .

The reliable test computes both $T\{x[\cdot - r]\}[n]$ and $(T\{x\})[n - r]$, keeping the system's own coefficients and index rule fixed.

Example 5.6 (Linear but time-varying). For $T\{x\}[n] = nx[n]$, the two expressions are $nx[n - r]$ and $(n - r)x[n - r]$. They differ, for example, for $x[n] = 1$, $r = 1$. For $T\{x\}[n] = x[n^2]$, they are $x[n^2 - r]$ and $x[(n - r)^2]$, also unequal for suitable inputs. Explicit dependence on the index does not imply nonlinearity.

Example 5.7 (Nonlinear but time-invariant). For $T\{x\}[n] = x[n]^2$, both paths give $x[n - r]^2$. Hence this nonlinear system is time-invariant. Any fixed memoryless rule $T\{x\}[n] = F(x[n])$ is time-invariant.

For the accumulator on $\mathcal{L}$, a change of variable proves time invariance:

$$A\{S_r x\}[n] = \sum_{k \leq n} x[k - r] = \sum_{m \leq n-r} x[m] = (Ax)[n - r].$$

Rate changers have the different shift laws in Proposition 4.7 and therefore fail the ordinary same-index time-invariance test.

5.4 Causality and memory

Definition 5.8 (Causality). The system T is causal if, for every integer N and every pair of inputs,

$$x_1[k] = x_2[k] \text{ for all } k \leq N \implies T\{x_1\}[N] = T\{x_2\}[N].$$

Equivalently, agreement of inputs through time N forces agreement of outputs through time N . A noncausal system violates this for at least one pair and time.

Definition 5.9 (Memory). The system is memoryless (static) if $x_1[N] = x_2[N]$ alone guarantees equal outputs at N , for every N . Equivalently, $T\{x\}[n] = F_n(x[n])$ for some functions F_n . Otherwise the input-output operator has memory (is dynamic).

Every memoryless system is causal, because agreement on the past includes agreement at the present sample. The converse is false: a one-sample delay is causal and has memory. An implementation can contain internal state that cancels out of the input-output relation; memory is classified from the operator, not merely from a particular circuit diagram.

Example 5.10 (How to prove noncausality). For $T\{x\}[n] = x[n] + x[n+1]$, choose $x_1 = 0$, $x_2 = \delta[n-1]$ and $N = 0$. The inputs agree through zero, but the outputs at zero are 0 and 1. For $x[n^2]$, use $N = 2$ and an impulse at 4. For $x[-n]$, use $N = -1$ and an impulse at 1. A single such example is decisive.

The accumulator is causal because its output at N uses only $k \leq N$. It has memory because inputs equal at N can differ at an earlier time and therefore have different accumulated sums. A *causal signal* simply means a sequence with $x[n] = 0$ for $n < 0$; it is a statement about support, not about a system processing an input.

5.5 Bounded-input bounded-output stability

Definition 5.11 (BIBO stability). A system is BIBO stable if every admissible bounded input produces a defined, bounded output:

$$\|x\|_\infty < \infty \implies \|Tx\|_\infty < \infty.$$

A proof usually bounds $|Tx[n]|$ uniformly in n using a bound on the input. One bounded input with an unbounded or undefined response disproves stability.

System	Proof or counterexample
$ax[n-r]$	$\ Tx\ _\infty \leq a \ x\ _\infty$: stable for finite a .
$x[n]^2$	$\ Tx\ _\infty \leq \ x\ _\infty^2$: stable.
$e^{x[n]}$	$ e^{x[n]} = e^{\operatorname{Re} x[n]} \leq e^{\ x\ _\infty}$: stable.
$x[g(n)]$	Selects input values, so $\ Tx\ _\infty \leq \ x\ _\infty$: stable.
$nx[n]$	$x[n] = 1$ gives $y[n] = n$: unstable.
Accumulator	$x[n] = u[n]$ gives $y[n] = (n+1)u[n]$: unstable.

The causal M -point moving average, $M \geq 1$, is

$$(H_M x)[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k]. \quad (12)$$

It is linear and time-invariant by distribution and substitution, causal by its indices, and stable since

$$|H_M x[n]| \leq \frac{1}{M} \sum_{k=0}^{M-1} |x[n-k]| \leq \|x\|_\infty.$$

For $M > 1$ it has finite memory; for $M = 1$ it is the memoryless identity.

5.6 Energy-based passivity

Definition 5.12 (Convention used here). A system is energy-nonincreasing (called passive in these lectures) if $\|Tx\|_2^2 \leq \|x\|_2^2$ for every finite-energy input. It is lossless if equality holds for every such input. This signal-energy convention is distinct from physical circuit passivity defined using supplied power and stored energy.

For $Tx[n] = ax[n-r]$, reindexing gives $E_{Tx} = |a|^2 E_x$. Hence it is passive exactly when $|a| \leq 1$, and lossless exactly when $|a| = 1$. Gain 2 is not passive, yet is BIBO stable; passivity and BIBO stability are different tests.

The moving average is passive: Cauchy–Schwarz gives

$$|H_M x[n]|^2 \leq \frac{1}{M} \sum_{k=0}^{M-1} |x[n-k]|^2.$$

Sum over n ; each shifted energy sum is E_x , giving $E_{H_M x} \leq E_x$. Proposition 4.7 similarly proves that zero insertion is lossless and sample deletion is energy-nonincreasing under the unweighted sequence-energy definition.

5.7 Impulse and step responses

Definition 5.13. Whenever the inputs belong to the system’s domain, define

$$h[n] = T\{\delta\}[n], \quad s[n] = T\{u\}[n].$$

These are responses to two particular test inputs; the definitions do not assume linearity or time invariance.

Direct substitution gives the following:

System	$h[n]$	$s[n]$
Identity	$\delta[n]$	$u[n]$
Accumulator	$u[n]$	$(n+1)u[n]$
Moving average H_M	$M^{-1} \sum_{k=0}^{M-1} \delta[n-k]$	$M^{-1} \sum_{k=0}^{M-1} u[n-k]$
Upsampler U_L	$\delta[n]$	$\sum_{k=0}^{\infty} \delta[n-kL]$
Downsampler D_M	$\delta[n]$	$u[n]$

For the three-point operator $Tx[n] = x[n] + \frac{1}{2}x[n-1] + \frac{1}{2}x[n+1]$,

$$h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \frac{1}{2}\delta[n+1],$$

$$s[n] = u[n] + \frac{1}{2}u[n-1] + \frac{1}{2}u[n+1] = \begin{cases} 0, & n \leq -2, \\ 1/2, & n = -1, \\ 3/2, & n = 0, \\ 2, & n \geq 1. \end{cases}$$

It is LTI and noncausal; its BIBO gain is at most $1 + 1/2 + 1/2 = 2$.

Precision check. The identity and every rate changer have the same $h[n] = \delta[n]$. Even knowing both h and s is insufficient for a general system: the identity and the nonlinear square operator agree on δ and u , but differ on 2δ . The convolution representation requires the LTI hypotheses and the convergence qualifications given next.

6 LTI systems and convolution

6.1 From impulse responses to an input-output formula

Definition 6.1 (LTI system). A system is linear time-invariant (LTI) if it satisfies both (10) and (11). Its response to δ is denoted by h .

Theorem 6.2 (Convolution representation for finite inputs). For every finite-support input to an LTI system,

$$T\{x\}[n] = \sum_{k \in \mathbb{Z}} x[k]h[n-k]. \quad (13)$$

Consequently, h determines the LTI system on all finite-support inputs.

Proof. Write $x = \sum_k x[k]S_k\delta$, a finite sum. By linearity and time invariance,

$$Tx = \sum_k x[k]TS_k\delta = \sum_k x[k]S_kT\delta = \sum_k x[k]S_kh.$$

Evaluating at n gives (13). $\square$

Infinite inputs require an extension argument. Let $x_N[k] = x[k]\mathbf{1}_{\{|k| \leq N\}}$. If the system preserves the relevant limit, so that $Tx_N[n] \rightarrow Tx[n]$, Theorem 6.2 yields the infinite convolution sum. For example, if $x \in \ell^1$, $T: \ell^1 \rightarrow \ell^\infty$ is LTI and bounded in the sense $\|Tz\|_\infty \leq C\|z\|_1$ for a fixed $C < \infty$, then $x_N \rightarrow x$ in ℓ^1 , hence $Tx_N \rightarrow Tx$ in ℓ^∞ . Also h is bounded, and $\sum_k |x[k]h[n-k]| \leq \|x\|_1 \|h\|_\infty$. Linearity alone justifies finite superposition, not arbitrary passage of an infinite sum through an operator.

Definition 6.3 (Linear convolution). When the sum converges, define

$$(x * h)[n] = \sum_{k \in \mathbb{Z}} x[k]h[n-k]. \quad (14)$$

All subsequent convolution-system claims refer to this defined operator on its stated domain. A convenient sufficient condition is $h \in \ell^1$, $x \in \ell^\infty$. Finite-support inputs also give finite sums; two right-sided sequences give finitely many contributing terms at each fixed output index.

The convolution operator is linear by distribution of the sum. It is time-invariant because

$$\sum_k x[k-r]h[n-k] = \sum_m x[m]h[(n-r)-m] = (x * h)[n-r],$$

whenever the reindexing is justified. Thus the usual convolution representation and LTI structure agree under these hypotheses.

6.2 The general linear, time-varying case

For a linear system, define the two-index impulse response

$$h_T[n, k] = T\{S_k\delta\}[n].$$

The same finite-superposition proof gives

$$Tx[n] = \sum_k x[k]h_T[n, k]$$

for finite-support inputs, with the preceding convergence qualification for infinite inputs. Time invariance gives the special form $h_T[n, k] = h[n-k]$. Conversely, that form implies time invariance on finite-support inputs and any compatible convolution extension. For example, the linear system $Tx[n] = nx[n]$ has $h_T[n, k] = n\delta[n-k]$, which depends on both indices; its response to the single impulse at zero is identically zero, although the system is not the zero operator.

6.3 Characterization of causality, memory and stability

Theorem 6.4 (LTI causality). A convolution system is causal exactly when

$$h[n] = 0 \quad \text{for every } n < 0.$$

Its causal-input response then simplifies to

$$y[n] = \sum_{k=0}^n x[k]h[n-k] \quad (n \geq 0), \quad y[n] = 0 \quad (n < 0). \quad (15)$$

Proof. If h vanishes at negative indices, $h[n-k] \neq 0$ requires $k \leq n$, so $y[n]$ uses no future inputs. Conversely, δ and the zero input agree through every $n < 0$. Causality and the zero response of a linear system imply $h[n] = 0$ there. If x is also causal, the additional condition $k \geq 0$ leaves exactly $0 \leq k \leq n$. $\square$

Proposition 6.5 (Memoryless LTI systems). A convolution system is memoryless exactly when $h[n] = c\delta[n]$ for a constant c . Its output is then $cx[n]$.

Proof. For any $n \neq 0$, the inputs δ and zero have the same value at n . Memorylessness forces their outputs there to coincide, so $h[n] = 0$. The remaining value $h[0] = c$ gives $(x * h)[n] = cx[n]$. This formula also proves sufficiency. $\square$

Theorem 6.6 (BIBO stability of a convolution system). The defining convolution series gives a bounded output for every bounded input if and only if

$$\boxed{h \in \ell^1, \quad \sum_{k \in \mathbb{Z}} |h[k]| < \infty.} \quad (16)$$

When this holds, the exact worst-case amplitude gain is

$$\sup_{\|x\|_\infty \leq 1} \|x * h\|_\infty = \|h\|_1.$$

Proof. If $|x[k]| \leq B$, absolute convergence and the triangle inequality give

$$|(x * h)[n]| \leq B \sum_k |h[n-k]| = B \|h\|_1,$$

uniformly in n . For necessity, choose the single bounded input

$$x[k] = \begin{cases} h^*[-k]/|h[-k]|, & h[-k] \neq 0, \\ 0, & h[-k] = 0. \end{cases}$$

Then $|x[k]| \leq 1$, but at $n = 0$,

$$(x * h)[0] = \sum_k x[k]h[-k] = \sum_k |h[-k]|.$$

If $h \notin \ell^1$, this nonnegative series diverges, so the convolution fails to provide a finite output for a bounded input. If $h \in \ell^1$, the same input attains the upper bound at zero, proving the exact gain formula. $\square$

For real coefficients the necessity input reduces to the sign of $h[-k]$. Boundedness of h alone is insufficient: the accumulator has $h = u$, which is bounded but not absolutely summable. Nor is square summability of h sufficient for BIBO stability on arbitrary bounded inputs.

6.4 Step response and its first difference

Proposition 6.7. If the step response of a convolution system exists, then

$$s[n] = (h * u)[n] = \sum_{k=-\infty}^n h[k], \quad \boxed{h[n] = s[n] - s[n-1]}. \quad (17)$$

Proof. The factor $u[n-k]$ in $\sum_k h[k]u[n-k]$ restricts the sum to $k \leq n$. Subtract consecutive cumulative sums. Alternatively, for any LTI system admitting the step input, $\delta = u - S_1 u$ directly gives $T\delta = Tu - S_1 Tu$ using only finite linearity and time invariance. $\square$

This formula need not hold for a non-LTI system. For example, it fails for an upsampler with $L > 1$, whose impulse and step responses were computed directly in Section 5.

6.5 Computing convolution and tracking support

For each fixed n , evaluate $\sum_k x[k]h[n-k]$ by folding $h[k]$ to $h[-k]$, shifting to $h[n-k]$, multiplying overlapping samples, and adding. The index k is the summation variable; n stays fixed during one such sum.

Proposition 6.8 (Support and overlap). If x is supported in $[a, b] \cap \mathbb{Z}$ and h in $[c, d] \cap \mathbb{Z}$, then

$$(x * h)[n] = \sum_{k=\max(a, n-d)}^{\min(b, n-c)} x[k]h[n-k], \quad (18)$$

with an empty sum equal to zero. The output is supported in $[a+c, b+d]$. If a, b, c, d are the actual nonzero endpoints, its span length is $N_x + N_h - 1$.

Proof. A contributing index must obey $a \leq k \leq b$ and $c \leq n-k \leq d$; intersecting these inequalities gives (18). No intersection is possible outside $[a+c, b+d]$. At the two endpoints, the unique terms are $x[a]h[c]$ and $x[b]h[d]$, which are nonzero. The span therefore has length $(b+d) - (a+c) + 1 = (b-a+1) + (d-c+1) - 1$. $\square$

Example 6.9 (Finite convolution). For $x = \{1, 2, 3\}$ and $h = \{1, 1, 1\}$,

n	0	1	2	3	4
Contributing sum	1	1+2	1+2+3	2+3	3
$y[n]$	1	3	6	5	3

Thus $y = \{1, 3, 6, 5, 3\}$. If the first input instead begins at index -1 and the second at index 2 , the same numerical list begins at output index 1 .

A tabular method lists the products $x[i]h[j]$ in a grid and adds those with $i+j=n$. Equivalently, multiply the finite formal polynomials $\sum_i x[i]z^{-i}$ and $\sum_j h[j]z^{-j}$; the coefficient of z^{-n} is $(x * h)[n]$. This is an algebraic computation device and requires no Z-transform theory.

Example 6.10 (Two causal exponentials). Let $x[n] = a^n$ and $h[n] = b^n$ for $n \geq 0$, zero otherwise, with the usual convention $a^0 = b^0 = 1$, including when a base is zero. Then $y[n] = 0$ for $n < 0$, and for $n \geq 0$,

$$y[n] = \sum_{k=0}^n a^k b^{n-k} = \begin{cases} (a^{n+1} - b^{n+1})/(a-b), & a \neq b, \\ (n+1)a^n, & a = b. \end{cases}$$

For $a \neq b$, multiply the sum by $a-b$; consecutive terms cancel. For $a = b$, all $n+1$ terms equal a^n . No infinite-series convergence condition is needed at a fixed n , since the causal overlap is finite.

6.6 Convolution identities with proofs

The following identities hold for finite-support sequences, for absolutely summable sequences, or under other hypotheses ensuring the required sums are absolutely convergent. For two right-sided sequences, the relevant fixed-index overlaps are finite. Do not freely rearrange conditionally convergent sums.

Proposition 6.11 (Algebra, shifts and reversal). If $y = x * h$, then

$$\begin{aligned} x * h &= h * x, & (x * h) * g &= x * (h * g), \\ x * (ah + bg) &= a(x * h) + b(x * g), & x * \delta &= x, \\ (ax) * h &= a(x * h), & x * (aS_r \delta) &= aS_r x, \\ (S_r x) * (S_s h) &= S_{r+s} y, & (Rx) * (Rh) &= Ry. \end{aligned}$$

In the scaled-impulse identity, equivalently $x * (a\delta[\cdot - r]) = ax[\cdot - r]$.

Proof. Commutativity: put $m = n - k$: $\sum_k x[k]h[n - k] = \sum_m h[m]x[n - m]$.

Associativity: substitute $r = k - i$ and exchange absolutely convergent sums:

$$\begin{aligned} ((x * h) * g)[n] &= \sum_k \sum_i x[i]h[k - i]g[n - k] \\ &= \sum_i x[i] \sum_r h[r]g[n - i - r] = (x * (h * g))[n]. \end{aligned}$$

When $x, h, g \in \ell^1$, the total absolute sum over all output indices and summation indices is at most $\|x\|_1 \|h\|_1 \|g\|_1$, which justifies these steps.

Distributivity and scaling: distribute the summand and pull out its constants. *Impulse:* the sole surviving term in $\sum_k x[k]\delta[n - k - r]$ has $k = n - r$; multiplying by a gives the scaled-impulse formula and setting $a = 1, r = 0$ gives the identity.

Shifts: with $i = k - r$,

$$\sum_k x[k - r]h[n - k - s] = \sum_i x[i]h[n - r - s - i] = y[n - r - s].$$

Reversal: with $i = -k$,

$$\sum_k x[-k]h[-n + k] = \sum_i x[i]h[-n - i] = y[-n].$$

□

In particular, reversing *both* inputs reverses the convolution. Reversing only one input does not generally do so. The full formula $y = x * h$ should be fixed before applying the shift identities.

6.7 Series and parallel interconnections

Proposition 6.12. Two convolution systems with responses h_1, h_2 have equivalent responses

$$h_{\text{cascade}} = h_1 * h_2, \quad h_{\text{parallel}} = h_1 + h_2,$$

where the parallel outputs are added. A gain a on one branch multiplies that branch's response by a .

Proof. A cascade gives $(x * h_1) * h_2 = x * (h_1 * h_2)$ by associativity. Added parallel outputs give $x * h_1 + x * h_2 = x * (h_1 + h_2)$ by distributivity. □

For LTI convolution systems, exchanging cascade order preserves the overall input-output relation by commutativity. This need not hold for arbitrary systems: applying gain 2 and then squaring gives $4x[n]^2$, while squaring and then applying gain 2 gives $2x[n]^2$.

If $h_1, h_2 \in \ell^1$, then

$$\|h_1 * h_2\|_1 \leq \sum_n \sum_k |h_1[k]| |h_2[n - k]| = \|h_1\|_1 \|h_2\|_1,$$

so a cascade of stable convolution systems is stable. A parallel sum is stable by the triangle inequality. Cascades and parallel sums of causal convolution systems remain causal by their nonnegative supports.

7 One-dimensional correlation

7.1 Definition, existence and relation to convolution

Definition 7.1 (Energy cross-correlation). For $x, y \in \ell^2$, define the aperiodic cross-correlation

$$r_{xy}[\ell] = \sum_{k \in \mathbb{Z}} x[k] y^*[k - \ell], \quad \ell \in \mathbb{Z}. \quad (19)$$

Autocorrelation is r_{xx} . For real sequences, conjugation has no effect. This is an uncentered, unnormalized inner-product similarity; it is not the Pearson correlation coefficient.

For every lag, Cauchy–Schwarz gives absolute convergence:

$$\sum_k |x[k] y^*[k - \ell]| \leq \left(\sum_k |x[k]|^2 \right)^{1/2} \left(\sum_k |y[k - \ell]|^2 \right)^{1/2} = \sqrt{E_x E_y}.$$

The energy convention is appropriate for finite records extended by zero and finite-energy sequences. Do not use the infinite energy sum for a persistent nonzero periodic signal; an average-based convention is given below.

Proposition 7.2 (Correlation as convolution). If $\tilde{y}[n] = y^*[-n]$, then $r_{xy} = x * \tilde{y}$.

Proof. $(x * \tilde{y})[\ell] = \sum_k x[k] \tilde{y}[\ell - k] = \sum_k x[k] y^*[k - \ell]$. □

In a direct lag-by-lag computation, correlate $x[k]$ with $y[k - \ell]$, conjugate the second factor, multiply and sum. To use a convolution routine, first reverse and conjugate y . Other books use the opposite lag sign; consistency of the definition determines the sign of the estimated delay.

7.2 Symmetry, energy and peak bounds

Theorem 7.3 (Correlation identities). For finite-energy sequences,

$$\begin{aligned} r_{xy}[\ell] &= r_{yx}^*[-\ell], & r_{xx}[0] &= E_x, \\ |r_{xy}[\ell]| &\leq \sqrt{E_x E_y}, & |r_{xx}[\ell]| &\leq r_{xx}[0], \\ r_{x\delta}[\ell] &= x[\ell], & r_{\delta x}[\ell] &= x^*[-\ell]. \end{aligned}$$

Correlation is linear in its first argument and conjugate-linear in its second:

$$r_{ax_1 + bx_2, y} = ar_{x_1 y} + br_{x_2 y}, \quad r_{x, ay_1 + by_2} = a^* r_{xy_1} + b^* r_{xy_2}.$$

If $x_r = S_r x$ and $y_s = S_s y$, then

$$r_{x_r y_s}[\ell] = r_{xy}[\ell - r + s]. \quad (20)$$

Proof. Conjugate $r_{yx}[-\ell]$ and set $m = k + \ell$:

$$r_{yx}^*[-\ell] = \sum_k y^*[k] x[k + \ell] = \sum_m x[m] y^*[m - \ell] = r_{xy}[\ell].$$

At zero lag $r_{xx}[0] = \sum_k |x[k]|^2$. Cauchy–Schwarz gives the cross-correlation bound, and setting $y = x$ gives the autocorrelation bound. The impulse identities follow by retaining the one nonzero term. Distributing the defining sum proves linearity in the first argument; conjugation of the second argument conjugates its scalars. Finally, put $m = k - r$:

$$\sum_k x[k - r] y^*[k - \ell - s] = \sum_m x[m] y^*[m - (\ell - r + s)].$$

□

Thus real autocorrelation is even, while complex autocorrelation is conjugate-symmetric. Cross-correlation need not be even and need not satisfy $r_{xy} = r_{yx}$.

If $y[n] = x[n - d]$ with $d > 0$, (20) gives

$$r_{xy}[\ell] = r_{xx}[\ell + d], \quad r_{yx}[\ell] = r_{xx}[\ell - d].$$

The maximum correlation magnitude is attained at $\ell = -d$ for r_{xy} , and at $\ell = d$ for r_{yx} . The sign follows from the convention, rather than from an informal rule that a delay must produce a positive lag.

7.3 A complete finite example and the lag range

Let $x = \{2, 1\}$ and $y = \{1, -1\}$. Their convolution is $x * y = \{2, -1, -1\}$, indexed from 0 through 2. Their correlation is instead

Lag ℓ	Nonzero terms in $\sum_k x[k]y[k - \ell]$	$r_{xy}[\ell]$
-1	$x[0]y[1] = 2(-1)$	-2
0	$x[0]y[0] + x[1]y[1] = 2 - 1$	1
1	$x[1]y[0] = 1$	1

Hence $r_{xy} = \{-2, 1, 1\}$, whereas $r_{yx} = \{1, 1, -2\}$. These satisfy the proved reversed-lag symmetry.

If $\text{supp } x \subseteq [a, b]$ and $\text{supp } y \subseteq [c, d]$, a contributing term requires $k \in [a, b] \cap [\ell + c, \ell + d]$. Therefore

$$r_{xy}[\ell] = 0 \quad \text{outside } [a - d, b - c].$$

For actual nonzero endpoints this span has length $N_x + N_y - 1$; some internal lags can still vanish. Explicit endpoint indices determine the origin of the reported correlation list.

7.4 Correlation for periodic or persistent signals

When appropriate limits exist, a time-average convention is

$$R_{xy}[\ell] = \lim_{K \rightarrow \infty} \frac{1}{2K + 1} \sum_{k=-K}^K x[k]y^*[k - \ell].$$

If x, y share a period N , the periodic-average proof used in (5) gives

$$R_{xy}[\ell] = \frac{1}{N} \sum_{k=0}^{N-1} x[k]y^*[k - \ell],$$

with periodic extension of the indices. In particular $R_{xx}[0] = P_x$. This is distinct from the unnormalized finite-energy correlation (19); state which convention a question uses.

8 Inverse systems and the role of the signal class

8.1 Invertibility and information loss

Definition 8.1 (Inverse on an admissible class). A system $T : \mathcal{X} \rightarrow \mathcal{Y}$ is invertible on $\mathcal{X}$ if there is $G : T(\mathcal{X}) \rightarrow \mathcal{X}$ such that $G(Tx) = x$ for every $x \in \mathcal{X}$. Thus G is defined on outputs actually produced by T . A two-sided inverse on all of $\mathcal{Y}$ additionally requires that T be onto $\mathcal{Y}$.

Theorem 8.2 (Injectivity criterion). An inverse on $T(\mathcal{X})$ exists exactly when T is injective. For a linear system this is equivalent to $\ker T = \{0\}$.

Proof. If $Tx_1 = Tx_2$, applying an inverse forces $x_1 = x_2$. Conversely, for an injective map each output in the range has exactly one preimage; map it to that preimage. For linear T , $Tx_1 = Tx_2$ is equivalent to $T(x_1 - x_2) = 0$, which proves the kernel criterion. $\square$

Squaring is not injective on real sequences because x and $-x$ have the same square; it is injective on nonnegative sequences. Downsampling by $M > 1$ loses information: 0 and $\delta[n-1]$ both map to zero. Upsampling is injective because $D_L U_L = I$, but $U_L D_L \neq I$ on arbitrary sequences: it zeros all indices not divisible by L .

8.2 Convolution inverses

Proposition 8.3. For convolution systems T_h, T_g , the condition for exact inverse cascading is

$$\boxed{h * g = \delta}, \quad (21)$$

provided the operations are defined on the chosen input/output classes and associativity is valid there.

Proof. If $h * g = \delta$, then $T_g(T_h x) = (x * h) * g = x * (h * g) = x$. Conversely, if the cascade is the identity, applying it to δ gives $h * g = \delta$. Commutativity supplies the reverse cascade whenever its domain and convergence conditions also hold. $\square$

The inverse response g is a convolution inverse, not the samplewise reciprocal $1/h[n]$. Existence of an inverse, existence of a *causal* inverse, and existence of a *stable* inverse are separate questions.

Example 8.4 (Gain and delay). For finite $a \neq 0$, $h[n] = a\delta[n-N]$ has inverse $g[n] = a^{-1}\delta[n+N]$. Their convolution is δ by the scaled-impulse identity. For $N > 0$, the original delay is causal but its exact inverse is an advance and is noncausal. Both responses are absolutely summable.

8.3 Accumulator and first difference: a two-sided calculation

On the left-finite class $\mathcal{L}$ of Example 5.2, define

$$(Ax)[n] = \sum_{k \leq n} x[k], \quad (\nabla x)[n] = x[n] - x[n-1].$$

Their impulse responses are $u[n]$ and $\delta[n] - \delta[n-1]$.

Proposition 8.5. On $\mathcal{L}$, $\nabla A = A\nabla = I$. The first difference is a causal, BIBO-stable inverse of the causal but unstable accumulator.

Proof. Adjacent partial sums cancel:

$$(\nabla Ax)[n] = \sum_{k \leq n} x[k] - \sum_{k \leq n-1} x[k] = x[n].$$

For the reverse order, choose a far enough in the zero prehistory that $x[a-1] = 0$. Then

$$(A\nabla x)[n] = \sum_{k=a}^n (x[k] - x[k-1]) = x[n] - x[a-1] = x[n].$$

Equivalently, $u * (\delta - S_1 \delta) = u - S_1 u = \delta$. The difference filter is causal and its impulse response has absolute sum two, so Theorem 6.6 gives stability. The bounded step input proves accumulator instability as before. $\square$

On unrestricted two-sided sequences, ∇ is not injective: it sends every constant sequence to zero. If merely $x[n] \rightarrow c$ as $n \rightarrow -\infty$, accumulating the first differences recovers $x[n] - c$, not $x[n]$. The boundary condition removes precisely this lost constant. Similarly, differentiating a running sum recovers its input even when the running sum has a nonzero additive constant, but that does not make the accumulator's initial state irrelevant to its input-output definition.

9 Difference equations and FIR/IIR filters

Scope. This short supplement develops topics introduced in the attached compiled notes. In the original W06_L02 lecture, the midsemester cutoff occurs after inverse systems and before the FIR/IIR heading.

9.1 Linear constant-coefficient difference equations

A causal recursive input-output description has the form

$$y[n] + \sum_{k=1}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]. \quad (22)$$

The coefficient of $y[n]$ has been normalized to one. To compute a unique response starting at a finite index, supply the needed past outputs and inputs. For the usual zero-state LTI operator, use zero prehistory and apply the same rule to shifted inputs with correspondingly shifted zero prehistory.

Proposition 9.1 (Zero-state linearity, time invariance and causality). On left-finite inputs, recurrence (22) with zero prehistory defines a causal LTI operator.

Proof. Starting before the first nonzero input, each $y[n]$ is uniquely determined by earlier outputs and present/past inputs, proving causality and uniqueness. A linear combination of two solutions satisfies the same recurrence driven by the corresponding linear combination of inputs, with zero prehistory; uniqueness proves linearity. Because the coefficients are constant, shifting a solution and its input gives a solution to the shifted problem, again with zero prehistory; uniqueness proves time invariance. $\square$

An imposed nonzero initial state adds a zero-input response and generally makes the map from input alone affine. For example,

$$y[n] = ay[n-1] + x[n], \quad y[-1] = c \quad \implies \quad y[n] = a^{n+1}c + \sum_{k=0}^n a^{n-k}x[k] \quad (n \geq 0).$$

This formula follows by repeated substitution or induction.

9.2 Reading and constructing a block realization

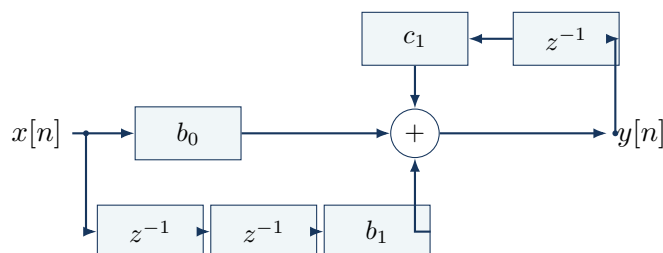
Rearrange (22) as

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k].$$

Each delay block z^{-1} stores one previous sample. Input taps with gains b_k are feedforward branches; delayed-output taps have feedback gains $-a_k$. Add all branch contributions at the output. For example, the lecture's form

$$y[n] = b_0 x[n] + b_1 x[n-2] + c_1 y[n-1]$$

uses a direct input tap, an input tap after two delays, and a feedback tap after one delay. Its normalized left-side coefficient is $a_1 = -c_1$.



9.3 Impulse-response definitions and stability

Definition 9.2. A convolution system is FIR (finite impulse response) if h has finite support, and IIR (infinite impulse response) if h has infinite support. These are properties of the impulse response, not definitions based on feedback in a diagram.

Every FIR system with finite coefficients is BIBO stable, because $\sum_n |h[n]|$ is a finite sum of finite numbers. An FIR system can be noncausal, for example $h[n] = \delta[n+1]$. An IIR system may be stable or unstable.

For a causal FIR system with $h[k] = b_k$, $0 \leq k \leq M$,

$$y[n] = \sum_{k=0}^M b_k x[n-k].$$

The moving average has $h[k] = 1/M$ for $k = 0, \dots, M-1$, zero otherwise, and the first difference has $h = \{1, -1\}$. Both are FIR. The accumulator has $h = u$, so it is IIR and unstable.

Proposition 9.3 (First-order recursion). For the zero-state system $y[n] = ay[n-1] + x[n]$,

$$h[n] = \begin{cases} a^n, & n \geq 0, \\ 0, & n < 0, \end{cases} \quad y[n] = \sum_{k=0}^n a^{n-k} x[k] \quad \text{for causal } x.$$

It is BIBO stable exactly when $|a| < 1$. It is IIR for $a \neq 0$ and FIR for $a = 0$, when it is the identity.

Proof. For an impulse input, $h[-1] = 0$, $h[0] = 1$, and $h[n] = ah[n-1]$ for $n \geq 1$. Induction gives a^n . Convolution yields the output formula. The stability test is $\sum_{n=0}^{\infty} |a|^n < \infty$, which holds exactly when $|a| < 1$. $\square$

Recursion does not imply IIR. Subtract consecutive moving-average outputs:

$$y[n] - y[n-1] = \frac{x[n] - x[n-M]}{M}.$$

This recursive realization still has the same finite impulse response when initialized consistently. An even simpler example is $y[n] - y[n-1] = x[n] - x[n-1]$: with zero prehistory, $y[n] - x[n]$ is a constant equal to zero, so the system is the FIR identity.

9.4 Why a symmetric FIR can have linear phase

The lecture lists linear-phase filtering as a possible DSP advantage. For a finite filter driven by $x[n] = e^{j\omega n}$, direct substitution gives

$$y[n] = e^{j\omega n} H(\omega), \quad H(\omega) = \sum_{k=0}^M h[k] e^{-j\omega k}.$$

If the coefficients are real and symmetric, $h[k] = h[M-k]$, then

$$H(\omega) = e^{-j\omega M/2} \underbrace{\sum_{k=0}^M h[k] e^{-j\omega(k-M/2)}}_{A(\omega) \text{ real}}.$$

Pairing terms k and $M-k$ cancels the imaginary parts and proves that $A(\omega)$ is real. Wherever $H \neq 0$, its phase is $-\omega M/2$ modulo a possible π jump when A changes sign. Thus the phase is $-\omega M/2$ plus a constant on each interval of nonzero constant sign. Antisymmetric real taps similarly give an imaginary factor and a constant $\pi/2$ phase offset. The claim applies to suitably chosen filters, not to every digital filter.

10 Reference tables and coverage

10.1 One-table system classification

The table uses two-sided sequence indices, fixed finite coefficients, and the zero-prehistory interpretation for recursive examples. Here TI means time-invariant.

System $y[n]$	Linear	TI	Causal	Memoryless	BIBO stable
$ax[n]$	yes	yes	yes	yes	yes
$x[n]^2$	no	yes	yes	yes	yes
$e^{x[n]}$	no	yes	yes	yes	yes
$nx[n]$	yes	no	yes	yes	no
$x[-n]$	yes	no	no	no	yes
$x[n^2]$	yes	no	no	no	yes
$x[n] + x[n-1]$	yes	yes	yes	no	yes
$x[n] + x[n+1]$	yes	yes	no	no	yes
Accumulator	yes	yes	yes	no	no
Moving average, $M > 1$	yes	yes	yes	no	yes
First difference	yes	yes	yes	no	yes
$U_L x$, $L > 1$	yes	no	no*	no	yes
$D_M x$, $M > 1$	yes	no	no*	no	yes

* Same-index causality on all two-sided sequences. The physical sampling clocks differ for rate changers; see Section 4. The accumulator is defined on inputs with convergent past sums, including the shift-closed class $\mathcal{L}$; its instability is witnessed by the admissible bounded input u .

10.2 Core formulae and proof methods

Signal, sampling and system identities

$$\begin{aligned}
 x[n] &= x_a(nT_s), & f_s &= 1/T_s, & \omega &= 2\pi f/f_s, \\
 \omega/(2\pi) = p/q \text{ in lowest terms} &\implies N_0 = q \text{ for a nonzero complex exponential,} \\
 f_2 &= \pm f_1 + kf_s \text{ for zero-phase cosine aliases,} & f_s &> 2f_{\max}, \\
 \Delta &= 2A/2^B, & |e_q| &\leq \Delta/2 \text{ without overload,} \\
 x[n] &= \sum_k x[k]\delta[n-k], & \delta[n] &= u[n] - u[n-1], \\
 (x * h)[n] &= \sum_k x[k]h[n-k], & s[n] &= \sum_{k \leq n} h[k], \\
 \text{causal convolution system} &\iff h[n] = 0 \ (n < 0), \\
 \text{BIBO-stable convolution system} &\iff h \in \ell^1, \\
 r_{xy}[\ell] &= \sum_k x[k]y^*[k-\ell], & r_{xy}[\ell] &= r_{yx}^*[-\ell], \\
 |r_{xy}[\ell]| &\leq \sqrt{E_x E_y}, & r_{xx}[0] &= E_x, \\
 h_{\text{cascade}} &= h_1 * h_2, & h_{\text{parallel}} &= h_1 + h_2, & h * g &= \delta.
 \end{aligned}$$

1. **Classify a system:** write the input class and initial conditions; test superposition, then compare $TS_r x$ with $S_r T x$. For causality use the input-pair definition; for stability establish a uniform-in-time bound or give a bounded counterexample.
2. **Manipulate a sequence:** mark its origin and support. Solve the index map rather than guessing a shift direction. Check the integer-index constraint.

3. **Compute convolution or correlation:** write the defining sum first, then determine overlap limits. Correlation reverses the lag when its arguments are exchanged and conjugates one argument for complex signals.
4. **Prove a general identity:** identify the index substitution. Check absolute convergence before exchanging infinite sums. A finite impulse expansion needs only linearity; an infinite one needs a justified extension.
5. **Discuss reconstruction or an inverse:** specify the admissible class. The sampling-rate condition supplies uniqueness only within its band-limited class; an inverse for a difference operator can depend on a boundary condition.

10.3 Source coverage and limits

Supplied source	Material retained	Location here
W01_L01; W01	Signals, classification, DSP motivation, conversion stages	Section 1
W02(1)	Sequence operations, impulse/step identities, parity, periodicity, energy/power, delays and block equations	Sections 2–3, 9
W03_04	Sampling/aliasing, rate changes, system properties, passivity, impulse/step responses and interpolation	Sections 4–5
W06_L01	LTI representation, convolution computation and properties, system interconnections	Section 6
W06_L02	Correlation, further convolution identities, inverse systems and midsemester cutoff	Sections 6–8
DSP_Midsem; DSP_Final_Notes (1)	Consolidated material, revision results and the short FIR/IIR introduction	Throughout; supplement in Section 9

The proofs and examples fill gaps within the covered material. The small spectral calculation for the sampling theorem and the finite-filter phase calculation are local justifications, not coverage of the full Fourier-series, CTFT, DTFT, DFT or Z-transform syllabus. Image and text processing are outside this notes set.